Questions-Gen — 系统架构 · 三阶段训练数据流

Training data → SFT → GRPO → DeepSeek distillation → validation → Hugging Face / Ollama

Primary flow
Fallback / feedback
Dashed = architecture zone
Box = module + responsibility

PROJECT OVERVIEW Questions-Gen is a Qwen3-14B system for generating and evaluating mathematical competition problems through staged training.
The pipeline combines supervised fine-tuning, GRPO reward optimization, DeepSeek-R1 distillation, validation, and local deployment.

TRAINING DATA · problem mixture

- **Historical Problems** — 50% source distribution
- **Variants** — 30% conditional variation
- **Novel Problems** — 20% innovation target

Stage 1 SFT · QuestionsGenTrainer.py

- **Qwen3-14B** — supervised fine-tuning
- **Stage-1 Checkpoint** — stage1-fp-merged

Stage 2 GRPO · reward optimization

- **RewardCalculator** — 0.4 difficulty · 0.3 novelty · 0.2 rigor · 0.1 diversity
- **NoveltyConstraint** — variation / duplication guard
- **GRPO Trainer** — group-relative optimization

Stage 3 Distill · teacher supervision

quality feedback

- **DeepSeekTeacher** — DeepSeek-R1 reasoning teacher
- **Final Model** — final-merged-16bit

VALIDATION

- **ModelValidator** — single checkpoint
- **BatchValidator** — algebra · geometry · calculus
- **QualityEvaluator** — clarity · difficulty · originality

DEPLOYMENT

- **Hugging Face** — three published stages
- **Ollama** — local inference
- **questions-gen CLI** — validate · batch · quality

主链：混合题库 → SFT → GRPO reward/novelty → DeepSeek-R1 distill → validate → publish。